

Haoxing Quant Assessment — LeetCode Prep Guide

Role: 策略部初级量化研究员 (Junior Quant Researcher, Strategy Department) **Company:** 上海皓兴信息技术有限公司 (Shanghai Haoxing) **Focus area:** Deep learning for intraday high-frequency signal research **Assessment:** LeetCode Enterprise, 90 minutes **Platform:** leetcode.cn

How to use this guide

- Problems are ordered by **priority** within each topic — do the starred (★) ones first.
 - Difficulty: E = Easy, M = Medium, H = Hard.
 - Target: solve 2–3 problems per day across different topics rather than grinding one topic to exhaustion.
 - Time-box each problem at 25 minutes. If stuck, read the editorial, understand, then re-implement from scratch.
 - All problem numbers are LeetCode numbers (work on both leetcode.com and leetcode.cn).
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Topic 1 — Stock / Trading DP (HIGHEST PRIORITY)

These are thematically aligned with the firm and teach the DP-with-state-machine pattern. Do all six.

#	Problem	Difficulty	Why
121	★ Best Time to Buy and Sell Stock	E	Foundation — one transaction
122	★ Best Time to Buy and Sell Stock II	M	Unlimited transactions
123	★ Best Time to Buy and Sell Stock III	H	At most 2 transactions — state machine kicks in
			k transactions —

188	Best Time to Buy and Sell Stock IV	H	generalization
309	★ Best Time to Buy and Sell Stock with Cooldown	M	Cooldown state (very on-theme)
714	★ Best Time to Buy and Sell Stock with Transaction Fee	M	Fee state

Topic 2 — Probability & Expected Value DP

High likelihood at quant firms. Your OR background makes these strong picks.

#	Problem	Difficulty	Why
688	★ Knight Probability in Chessboard	M	Classic expected-value DP
837	★ New 21 Game	M	Stopping-time probability
808	Soup Servings	M	Expected value with memoization
1227	Airplane Seat Assignment Probability	M	Elegant probability reasoning
1230	Toss Strange Coins	M	DP over probability distributions
790	Domino and Tromino Tiling	M	Counting / probability-adjacent DP
935	Knight Dialer	M	Counting paths

Custom problem to practice (from earlier in our chat):

- Roll a fair die until running sum > N. Compute expected final sum. O(N) DP.

Topic 3 — Sliding Window & Two Pointers

Warm-up problem type. Bank these quickly on test day.

#	Problem	Difficulty	Why
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3	★ Longest Substring Without Repeating Characters	M	Classic sliding window
76	Minimum Window Substring	H	Hard but canonical
209	★ Minimum Size Subarray Sum	M	Sliding window with sum
424	Longest Repeating Character Replacement	M	Window + frequency
567	Permutation in String	M	Window + frequency compare
1004	Max Consecutive Ones III	M	Good warm-up
1438	★ Longest Continuous Subarray with Absolute Diff $\leq$ Limit	M	Monotonic deque
239	Sliding Window Maximum	H	Monotonic deque classic
11	Container With Most Water	M	Two pointers
15	3Sum	M	Two pointers + dedup

Topic 4 — Hash Maps & Prefix Sums

#	Problem	Difficulty	Why
1	★ Two Sum	E	Warm-up
560	★ Subarray Sum Equals K	M	Prefix sum + hash
974	Subarray Sums Divisible by K	M	Prefix sum + modular
525	Contiguous Array	M	Prefix sum trick
49	Group Anagrams	M	Hashing strategy
128	Longest Consecutive Sequence	M	Hash set
146	LRU Cache	M	Hash + doubly linked list

Topic 5 — Heaps / Priority Queues

#	Problem	Difficulty	Why
215	★ Kth Largest Element in an Array	M	Heap or quickselect
347	★ Top K Frequent Elements	M	Heap + hash
295	Find Median from Data Stream	H	Two heaps — common quant problem
253	Meeting Rooms II	M	Heap for intervals
1046	Last Stone Weight	E	Heap basics
23	Merge k Sorted Lists	H	Heap of heads

Topic 6 — Binary Search (including “binary search on the answer”)

#	Problem	Difficulty	Why
704	Binary Search	E	Foundation
33	★ Search in Rotated Sorted Array	M	Classic variant
34	Find First and Last Position	M	Lower/upper bound
153	Find Minimum in Rotated Sorted Array	M	Canonical
875	★ Koko Eating Bananas	M	Binary search on the answer
1011	★ Capacity To Ship Packages Within D Days	M	Binary search on answer
410	Split Array Largest Sum	H	Advanced binary search on answer
4	Median of Two Sorted Arrays	H	Hard but classic

Topic 7 — Dynamic Programming (General)

#	Problem	Difficulty	Why
70	Climbing Stairs	E	DP foundation
198	★ House Robber	M	1D DP
213	House Robber II	M	Circular variant
300	★ Longest Increasing Subsequence	M	$O(n^2)$ then $O(n \log n)$
53	★ Maximum Subarray	M	Kadane's algorithm
152	Maximum Product Subarray	M	State-machine DP
322	Coin Change	M	Unbounded knapsack
416	Partition Equal Subset Sum	M	0/1 knapsack
1143	Longest Common Subsequence	M	2D DP foundation
72	Edit Distance	M	2D DP classic
647	Palindromic Substrings	M	Interval DP

Topic 8 — Monotonic Stack

#	Problem	Difficulty	Why
496	Next Greater Element I	E	Foundation
739	★ Daily Temperatures	M	Canonical monotonic stack
84	Largest Rectangle in Histogram	H	Classic hard
42	★ Trapping Rain Water	H	Two approaches: stack or two pointers
901	Online Stock Span	M	Thematic for trading

Topic 9 — Math / Numerical

#	Problem	Difficulty	Why
50	★ Pow(x, n)	M	Fast exponentiation
372	Super Pow	M	Modular exponentiation
69	Sqrt(x)	E	Newton's method or binary search
172	Factorial Trailing Zeroes	M	Math insight
470	Implement Rand10() Using Rand7()	M	Probability-flavored
528	Random Pick with Weight	M	Prefix sum + binary search
398	Random Pick Index	M	Reservoir sampling
382	Linked List Random Node	M	Reservoir sampling

Topic 10 — Graphs (lower priority, but basics matter)

#	Problem	Difficulty	Why
200	★ Number of Islands	M	BFS/DFS foundation
207	Course Schedule	M	Topological sort
743	Network Delay Time	M	Dijkstra
787	Cheapest Flights Within K Stops	M	Bellman-Ford variant
399	Evaluate Division	M	Graph + DFS (arbitrage-adjacent)

Topic 11 — ML From Scratch (possible for this specific role)

Not LeetCode problems but important given the deep learning focus of the role. Practice implementing these in NumPy, no frameworks:

- **Softmax + cross-entropy loss** (forward and backward pass)
- **2-layer MLP forward pass** with ReLU activation
- **Single-head self-attention** for a small sequence (Q, K, V, scaled dot product)
- **Batch normalization** forward pass
- **Simple 1D convolution** for time series
- **Gradient of sigmoid and tanh** (derive by hand, then code)

Resources: Stanford CS231n notes; Andrej Karpathy's micrograd for the grad-from-scratch mindset.

Topic 12 — Probability / Statistics Concepts (for any multiple-choice section)

Refresh these before the test:

- **Expected value & variance** — linearity of expectation, $\text{Var}(X+Y)$ when independent
 - **Conditional probability** — Bayes' rule, law of total probability
 - **Common distributions** — Bernoulli, Binomial, Geometric, Poisson, Normal, Exponential; know mean and variance of each
 - **Central limit theorem** — when it applies, what it says
 - **Correlation vs covariance** — definitions and relationship
 - **Hypothesis testing basics** — null hypothesis, p-value, Type I/II errors (less critical but good to know)
 - **Stationarity and autocorrelation** — relevant to time-series / high-frequency signal work
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Topic 13 — Time Series & Signal Processing (role-specific)

Concepts that may appear in multiple-choice or discussion:

- **Moving averages** — simple, weighted, exponential (EWMA)
- **Autocorrelation function** — interpretation

- **Stationarity** — why it matters in financial time series
 - **Log returns vs simple returns** — when to use which
 - **Sharpe ratio, information ratio** — formulas and intuition
 - **Backtesting pitfalls** — look-ahead bias, survivorship bias, overfitting
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Suggested 7-day prep schedule

Day 1 — Stock/Trading DP Problems 121, 122, 309, 714 (all six if time permits).

Day 2 — Probability DP Problems 688, 837, 808. Plus the “roll until sum > N” custom problem.

Day 3 — Sliding window & two pointers Problems 3, 209, 1438, 11, 15. Bank these fast.

Day 4 — Hash maps, prefix sums, heaps Problems 560, 215, 347, 295. Mix of fast wins and one harder one.

Day 5 — Binary search & general DP Problems 875, 1011, 198, 300, 53.

Day 6 — Math, monotonic stack, graph basics Problems 50, 739, 42, 200.

Day 7 — ML from scratch + review Write softmax+cross-entropy, a 2-layer MLP forward pass, and a simple attention block in NumPy. Then re-review 5–10 problems from the week that felt shaky.

Day-of-test checklist

- Use **Chrome** (required per the invitation email).
- Close all other LeetCode tabs and AI assistants (Copilot, ChatGPT, etc.).
- Open a blank note / scratch pad for working out math or sketching approaches.
- Have a Chinese–English translation tab ready (DeepL or Google Translate).
- Read all problems first (5 min), solve easiest first, lock in correctness before optimizing.
- Write your own test cases (empty input, single element, duplicates, negatives, overflow) before submitting.
- If stuck 20+ minutes on one problem, move on and come back.

Final notes

- **Language recommendation:** Python. Less boilerplate, `collections.deque`, `heapq`, `Counter`, `defaultdict` all built in. Unless the role explicitly tests C++, use Python.
- **Partial credit matters.** Chinese firms generally weigh correctness highly — a fully-solved medium beats two half-solved hard.
- **Don't panic on the probability questions.** Given your background in stochastic modeling and operations research, these should be your advantage. Read carefully, set up the recurrence, code it up.

Good luck — 加油!